

Sample Programs for Practice

1) Write a single program to generate a multiplication and subtraction table for a given number.

2) Write a program to print a letters from the user input character to 'Z' without using strings.

Example 1: input : X
Output : XYZ

Example 2: Input : M
Output : MNOPQRSTUVWXYZ

3) write a program to find the difference between the given number and it's reverse number.

Example 1: Input : 12345
Output : 12345 - 54321 = -41976

Example 2: Input : 543
Output : 543 - 345 = 198

4) Write a program to define the functionality of a Bitwise operators.

5) Write a program to print the following pattern based on the alphabet.

Example 1:

Input: E
Output:
EEEE
DDDD
CCC
BB
A

Example 2:

Input:C

Output:

CCC
BB
A

6) Write a program for the following. Let consider 20 students in a class with the roll no 101-120. We are going to split them into four groups. Based on their roll number we are going to split them with the following logics.

Example 1:

Input : Total Number of students : 20

Output:

Group 1:

101
105
109
113
117

Group 2:

102
106
110
114
118

Group 3:

103
107
111
115
119

Group 4:

104
108
112
116

7) Write a program to print the following output for the given input. You can assume the string is of odd length

Eg 1: Input: 12345

Output:

```
1  1
2 2
3
4 4
5  5
```

8) Write a program to demonstrate the functionalities of static keyword.

9) Given two sorted arrays, merge them such that the elements are not repeated

Example 1:

Input: Array 1: 2,4,5,6,7,9,10,13

Array 2: 2,3,4,5,6,7,8,9,11,15

Output: Merged array: 2,3,4,5,6,7,8,9,10,11,13,15

10) Alternate sorting: Given an array of integers, rearrange the array in such a way that the first element is first maximum and second element is first minimum.

Example: Input : {1, 2, 3, 4, 5, 6, 7}

Output: {7, 1, 6, 2, 5, 3, 4}

11) Java Program to left rotate the elements of a multidimensional array.

12) Write a program to print the number 7.50 without using String.

13) Write a program to find the area of the square

14) Write a program to find the area of the rectangle

15) Write a program to find the area of the cylinder

16) Write a program to evaluate the polynomial equation

$$aX^2 + bX + c = 0$$

$$X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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17) Given the radius of two concentric circle, find the area of the space between the circles.

18) Evaluate the following expression on paper and on program and understand the difference

i) ++a-b- ii) a%b++ iii) a*=b+5 iv) x=69>>>2

19) Evaluate the following expression on paper and on program and understand the difference

a+=a++ + ++a + -a + a-; when a=28

20) Evaluate the following expression on paper and on program and understand the difference

x = x++ * 2 + 3 * -x;

21) Evaluate the following expression on paper and on program and understand the difference

If int y = 10 then find int z = (++y * (y++ + 5));

22) Evaluate the following expression on paper and on program and understand the difference

23) What is the value of x1 if x=5 ?

x1=++x - x++ + -x

- 24) Program to Find ASCII Value of a Character
- 25) Program to Compute Quotient and Remainder
- 26) Program to Demonstrate the Working of Keyword long
- 27) Program to Swap Two Numbers
- 28) Program to Check Whether a Number is Even or Odd
- 29) Program to Check Whether a Character is a Vowel or Consonant
- 30) Program to Find the Largest Number Among Three Numbers
- 31) Program to Find the Roots of a Quadratic Equation
- 32) Program to Check Whether a Number is Positive or Negative
- 33) Program to Check Whether a Character is an Alphabet or not
- 34) Program to Calculate the Sum of Natural Numbers and find the largest digit of the sum.
- 35) Program to Find Factorial of a Number and the factorial number's sum of digits.
- 36) Java Program to print the elements of an array present on even position
- 37) Java Program to print the elements of an array present on odd position
- 38) Java Program to print the largest element in an array
- 39) Java Program to print the smallest element in an array
- 40) Java Program to print the number of elements present in an array
- 41) Java Program to copy all elements of one array into another array
- 42) Java Program to find the frequency of each element in the array
- 43) Java Program to left rotate the elements of an array
- 44) Java Program to print the duplicate elements of an array
- 45) Java Program to print the sum of all the items of the array
- 46) Java Program to rotate the elements of an array in the right direction
- 47 - 50) Write a programs to print following patterns

```
      1
    1 2 1
  1 2 3 2 1
1 2 3 4 3 2 1
  1 2 3 4 5 4 3 2 1
    1 2 3 4 5 6 5 4 3 2 1
      1 2 3 4 5 6 7 6 5 4 3 2 1
        1 2 3 4 5 6 7 8 7 6 5 4 3 2 1
```

```
      9
    8 9 8
  7 8 9 8 7
6 7 8 9 8 7 6
5 6 7 8 9 8 7 6 5
4 5 6 7 8 9 8 7 6 5 4
3 4 5 6 7 8 9 8 7 6 5 4 3
2 3 4 5 6 7 8 9 8 7 6 5 4 3 2
1 2 3 4 5 6 7 8 9 8 7 6 5 4 3 2 1
```

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * *
* * * *
* * *
* *
*

```

```

9 9 9 9 9 9 9 9 9
8 8 8 8 8 8 8 8
7 7 7 7 7 7 7
6 6 6 6 6 6
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1

```

51) Write a program to display the number in reverse order without use of String functions.

Eg 1: Input: 12345

Output :The number in reverse order is : 54321

52) Using Recursion reverse the string such as

Eg 1: Input: one two three

Output: three two one

Eg 2: Input: I love india

Output: india love I

53) Write a program to give the following output for the given input:

Eg 1: Input: a1b10

Output: abbbbbbbbbbb

Eg: 2: Input: b3c6d15

Output: bbbccccccddddd

The number varies from 1 to 99.

54) Write a program to sort the elements in odd positions in descending order and elements in ascending order.

Eg 1: Input: 13,2 4,15,12,10,5

Output: 13,2,12,10,5,15,4

Eg 2: Input: 1,2,3,4,5,6,7,8,9

Output: 9,2,7,4,5,6,3,8,1

55) Write a program to print the alphabet 'C' with the stars based on the given numbers

Eg 1: Input: 4

Output:

*

*

56) Write a program to make such a pattern like a pyramid with numbers increased by 1.

Eg: Input : 10

Output:

1

2 3

4 5 6

7 8 9 10

57) Write a program to print the following output for the given input. You can assume the string is of odd length

Eg 1: Input: 12345

Output:

1 5

2 4

3

2 4

1 5

58) Write a program to print the following pattern for the given input number.

Eg 1: Input: 4

Output:

*** _ ***

** _ **

* _ *

** _ **

*** _ ***

59) Find if a String2 is substring of String1. If it is, return the index of the first occurrence. else return -1.

Eg 1: Input: String 1: test123string

String 2: 123

Output: 4

Eg 2: Input: String 1: testing12

String 2: 1234

Output: -1

60) Write a program to find the sum of the series. The series will be like 1 +11 + 111 + 1111 +.. n terms.

Eg 1: Input: Input the number of terms : 5

Output :

1 + 11 + 111 + 1111 + 11111

The Sum is : 12345

Eg 2: Input: Input the number of terms : 3

Output :

1 + 11 + 111

The Sum is : 123

Programs on Arrays and Strings: Write minimum 2 different versions of java programs for each question.

61. Java Program to print the elements of an array in reverse order

62. Java Program to print the elements of an array present on even position and odd positions

63. Java Program to print the largest element and smallest element in an array

64. Java Program to copy all elements of one array into another array

65. Java Program to find the frequency of each element in the array

66. Java Program to left rotate the elements of an array

67. Java Program to print the duplicate elements of an array

68. Java Program to print the sum of all the items of the array

69. Java Program to rotate the elements of an array in the right direction

70. Java Program to find the 2nd Largest Number in an Array

71. Java Program to find the 2nd Smallest Number in an Array

72. Java Program to remove Duplicate Element in an Array
73. Java Program to add two matrices
74. Java Program to find the product of two matrices
75. Print Odd and Even Number from an Array
76. Java Program to display the transpose matrix
77. Java Program to sort the elements of an array in ascending order
78. Java Program to subtract the two matrices
79. Java Program to determine whether a given matrix is an identity matrix
80. Java Program to determine whether a given matrix is a sparse matrix
81. Java Program to determine whether two matrices are equal
82. Java Program to display the lower triangular matrix
83. Java Program to display the upper triangular matrix
84. Java Program to find the frequency of odd & even numbers in the given matrix
85. Java Program to find the sum of each row and each column of a matrix
86. Java Program to print all the characters in an arrays.
87. Java Program to convert char array to string without using toString() method.
88. Java Program to convert a sentence/paragraph to word(String) array with using string function.
89. Java Program to convert a sentence/paragraph to word(String) array without using string function.
90. Java Program to sort the words in ascending.
91. Java Program to sort the characters in a word in descending.
92. Java Program to sort the words in ascending using recursion.
93. Java Program to sort the characters in a word in descending using recursion.
94. Java Program to find the given two strings are anagram to each other
95. Java Program to find the given two strings are palindrome.
96. Java program to find a character in a word string linear search.
97. Java program to find a character in a given string using binary search.
98. Java Program to print the print the equivalent Capital letter of a given small letter in Alphabets.
99. Java Program to convert all small letter to capital letter without using special functions.
100. Java Program to convert ASCII value to character(for A-Z only) without using special function